**ITEM 1:** A renewable energy company is producing bioethanol from sugarcane molasses. The bioethanol is used to power rural micro-grids. The process involves fermenting glucose into ethanol and carbon dioxide using yeast, followed by fractional distillation. The following equilibria and thermochemical data are available:

| **Substance** | **Equilibrium concentration (mol/L) at 37 °C** |
|---|---|
| Glucose | 0.20 |
| Ethanol | 0.30 |
| Carbon dioxide | 0.30 |

To optimise yeast activity, the pH must be kept between 4.5 and 5.5. The team proposes using the acetic acid ($CH_3COOH$) / sodium acetate ($CH_3COONa$) buffer system (pKa = 4.74), prepared with 0.1 M acetic acid and 8.4 g sodium acetate in a total volume of 1.0 L.

However, current production suffers from low yield, energy inefficiency, and fluctuating reaction rates due to changing environmental conditions. Using 1.5 g of glucose as your starting feedstock, you are required to analyse the process and recommend improvements.

The table below provides additional kinetic data for glucose fermentation:

| **[Glucose] (mol/L)** | **[Yeast] (g/L)** | **Initial Rate (mol/L·min)** |
|---|---|---|
| 0.10 | 1 | 0.003 |
| 0.20 | 1 | 0.006 |
| 0.20 | 2 | 0.012 |
| 0.20 | 3 | 0.018 |

The management of the company has asked your school science club to carry out a pilot project to improve the yield, efficiency, and consistency of the fermentation process.

**Task**

As a member of the science club and chemistry student:

a. By applying the equilibrium law, predict how continuous removal of $CO_2$ would influence the ethanol yield.

b. Determine the suitability of the proposed buffer system to maintain optimal yeast activity during fermentation

c. From the kinetic data, determine the rate constant for fermentation and suggest scientifically sound methods to increase the rate constant.

d. Suggest ways by which the community members can utilise your findings to manage the fermentation process for their own benefit.

**ITEM 2:** Chemical industries producing dyes, pharmaceuticals, and other products rely on solvents like naphthalene and precise knowledge of molecular characteristics of additives and impurities. At the same time, the local water supply shows signs of contamination, with an unknown solute T affecting the freezing point of water in storage tanks.

The water treatment facility collected the following data on solute T:

| Concentration of T (g $dm^{-3}$) | 0 | 30 | 60 | 90 | 120 | 150 |
|---|---|---|---|---|---|---|
| Freezing point (°C) | 0 | -0.16 | -0.32 | -0.49 | -0.65 | -0.81 |

The Cryoscopic constant for water is 1.86 °C per 1000g per mole

The same company is optimizing an industrial reaction:

$$A+2B \rightleftharpoons 2C+D \qquad \Delta H^{\theta}=-50\ kJ\ mol^{-1}$$

| Exp No. | [A] (mol/L) | [B] (mol/L) | Initial rate (mol $L^{-1}$ $min^{-1}$) |
|---|---|---|---|
| I | $4.0\times10^{-2}$ | $4.0\times10^{-2}$ | $6.40\times10^{-3}$ |
| II | $4.0\times10^{-2}$ | $8.0\times10^{-2}$ | $1.28\times10^{-2}$ |
| III | $8.0\times10^{-2}$ | $4.0\times10^{-2}$ | $2.56\times10^{-2}$ |
| IV | $8.0\times10^{-2}$ | $8.0\times10^{-2}$ | $5.12\times10^{-2}$ |

The community depends on safe drinking water and efficient chemical production for economic growth. Your chemistry teacher has requested you to analyze the information to help solve these community problems.

Tasks

a. Construct a rate equation for the industrial reaction
b. Predict the behaviour of the reaction under different conditions of temperature and pressure
c. Determine the molecular mass of T so that the chemical company can accurately identify additives, ensure product quality and solve the water quality problem

## SECTION B

### Part I: Attempt any one item in this part

**ITEM 3:** Uganda faces persistent challenges in rural electrification. Many communities rely on unreliable and costly energy sources like diesel generators, kerosene lamps, and car batteries. A Ugandan company is developing solar-powered micro-grids and has invited your chemistry class to help identify suitable materials for solar batteries and wiring. These materials must be affordable, corrosion-resistant, and durable under Uganda's rural conditions.

The company is investigating elements similar to those in Periods 2 and 3 of the Periodic Table. The measured properties of selected elements are shown below:

| Element | Atomic Radius (pm) | First Ionization Energy (kJ/mol) | Electronegativity (Pauling) | Typical Bonding Type |
|---|---|---|---|---|
| Li | 145 | 520 | 1.0 | Metallic |
| Mg | 130 | 730 | 1.5 | Metallic |
| Al | 110 | 1000 | 2.5 | Metallic |
| Si | 95 | 1250 | 3.0 | Covalent |
| P | 85 | 1500 | 3.5 | Covalent |

Task

As a chemistry learner;

Use your understanding of atomic structure, periodic trends, chemical bonding and the information provided to propose suitable materials for solar batteries and wiring in Uganda's rural micro-grid systems

**ITEM 4:** Your chemistry class has been invited by a chemical plant management near Lake Victoria to assist in identifying ways of optimizing the production of key industrial compounds including sodium oxide, magnesium oxide and silicon dioxide used in glassmaking and ceramics. The plant management has observed inconsistencies in melting points and reactivity, which are affecting product quality and safety. You are tasked with conducting a scientific investigation to analyze periodic trends of the elements, compound properties, and molecular structures to recommend improvements.

Important chemical data of the findings about the elements and their compounds is provided in the table below to assist in the analysis.

| **Element** | **Atomic Number** | **Atomic Radius (pm)** | **Ionisation Energy (kJ/mol)** | **Melting Point (°C)** | **Oxide Melting Point (°C)** |
|---|---|---|---|---|---|
| Sodium (Na) | 11 | 186 | 496 | 98 | 1275 |
| Magnesium (Mg) | 12 | 160 | 738 | 650 | 2800 |
| Aluminium (Al) | 13 | 143 | 578 | 660 | 2072 |
| Silicon (Si) | 14 | 118 | 786 | 1410 | 1710 |
| Phosphorus (P) | 15 | 110 | 1012 | 44 | 580 |
| Sulphur (S) | 16 | 104 | 1000 | 115 | Gas/Sublimes |
| Chlorine (Cl) | 17 | 99 | 1251 | -101 | Gas/Gas |

**Task**

As a chemistry learner, write a report about the periodic trends of the elements and their oxides, and recommend improvements to the plant's production processes.

**Part II: Attempt any one item in this part**

**ITEM 5:** A certain town in central Uganda faces growing concerns over water pollution from agro-processing industries, especially during the cocoa harvest season. Wastewater from cocoa processing plants often contains organic matter including acids, and amines that can harm aquatic life if released untreated.

A group of entrepreneurs is setting up a small plant to process cocoa husks, a by-product of cocoa bean production, into value-added products. Chemical analysis of the husks shows that they contain ethanol (from natural fermentation), ethanoic acid, small amounts of amines (from protein breakdown), and aromatic aldehydes such as vanillin(4-hydroxy-3-methoxybenzaldehyde).

To address the water pollution challenge and create additional revenue streams, the team plans to:

1. Determine the relationship between the structure of vanillin and ethanoic acid, and their solubility.
2. Convert ethanol into amines for use in water treatment and cosmetics.
3. React ethanoic acid with ethanol to produce an ester for fragrances, and study its reaction mechanism.
4. Evaluate the use of the esters and amines in producing fragrances and cosmetics

You have been tasked by your teacher to design feasible chemical processes, explain the underlying organic chemistry, and propose sustainable solutions to help the plant reduce environmental harm while maintaining profitability.

**Tasks**

As a chemistry learner, address the plans for the team and propose suitable solutions.

**ITEM 6:** In southwestern Uganda, a small cosmetics start-up is producing herbal skin-care creams using locally sourced plant oils, such as shea butter and sunflower oil. The production process generates significant amounts of waste plant oils and fats. These wastes are currently disposed of into nearby drainage systems, causing blockages and foul smells, which has led to complaints from the community and environmental authorities.

The company has approached a team of A-level chemistry students in your school to:

- Analyse the composition of these waste materials, their functional groups and physical properties,

- Explore possible chemical processes to convert them into valuable products and their mechanisms,
- Design a synthetic route to convert compound C into compound D
- Propose a sustainable chemical process to convert waste A into a useful, marketable product that reduces environmental harm.

Analysis of a sample of the waste oil revealed the presence of a mixture of the following compounds:

| **Compound** | **Structure (condensed)** |
|---|---|
| A | $CH_3(CH_2)_{14}COOCH_2CH(OH)CH_2OH$ |
| B | $CH_3(CH_2)_7CH=CH(CH_2)_7COOH$ |
| C | $CH_3CH_2CH_2OH$ |
| D | $CH_3COCH_3$ |

**Tasks:**

As a chemistry learner, give written presentation that addresses the company's challenges and outlines scientifically sound, sustainable solutions.

# SET 2 - SAMPLE ITEMS FOR CHEMISTRY

**ITEM 1:** In several rural Ugandan universities, students rely on cheap portable power banks to charge their phones due to unreliable electricity. Recently, some of these power banks have overheated, swelled, and even caught fire in dormitories—causing panic among parents and school administrators.

Your chemistry class has been invited to work with a team of investigators aiming to find solutions for the problem. Scientific investigations revealed that these power banks are assembled using a mix of zinc-carbon cells and discarded lithium-ion cells, charged by unregulated solar panels. Inside, zinc acts as the anode and copper wires as the cathode, forming uncontrolled galvanic cells. Regulators for these setups are not available locally.

The chemistry team collected the following data to understand the issue:

| Standard electrode potentials: | Thermochemical data: | Additional data: |
|---|---|---|
| $Zn^{2+}/Zn$ = -0.76 V<br>$Cu^{2+}/Cu$ = +0.34 V<br>$Li^{+}/Li$ = -3.04 V<br>$Fe^{2+}/Fe$ = -0.44 V | • Enthalpy of combustion of lithium = -598 kJ/mol<br>• Enthalpy of neutralisation (NaOH + HCl) = -57 kJ/mol<br>• Specific heat capacity of water = 4.18 J/g·K<br>• 80 g of leaked fluid rose from 25 °C to 60 °C during a fire | • Faraday's constant = 96,500 C/mol<br>• Charging current = 2 A for 30 minutes<br>• Recommended safer alternative: iron-nickel cells ($Fe^{2+}/Fe$ = -0.44 V; $Ni^{2+}/Ni$ = -0.25 V) |

**Tasks**

As a member of the chemistry class,

a) Explain using data provided how spontaneous redox reactions in the power banks can cause overheating and fires.

b) Assess whether iron-nickel batteries are a safer alternative for rural school

**ITEM 2**: You are a chemical engineer at a fertilizer production plant that manufactures Ammonium nitrate by the reaction of Ammonia ($NH_3$) and Nitric acid.

Ammonia is produced as a product of the Haber process with Kc = 0.16 at 500 K, ΔH = −92 kJ/mol.

The nitrogen used has isotopic abundances of 99.63% ${}^{14}_{7}N$ and 0.37% ${}^{15}_{7}N$. The plant aims to produce 1000 kg of Ammonium nitrate daily (O=16, H=1). Your employer has asked you to give ideas on the following:

- Constructing an energy profile diagram for the manufacture of Ammonia.

- Determining the moles of Ammonia and Nitric acid required to produce the daily amount of Ammonium nitrate and the volume of 2.0 M Nitric acid solution needed.
- Calculating the equilibrium concentrations starting with $[N_2] = 0.5$ M, $[H_2] = 1.5$ M, and $[NH_3] = 0$ M at 500 K.
- Devising a means of optimizing the process by while ensuring efficient use of resources.

**Task**

Make write up to address the concerns of the employer

**SECTION B**

**Part I: Attempt any one item**

**ITEM 3:** You are a chemistry student intern at a local environmental and health laboratory. Communities near several farms have reported to the laboratory management that wells and streams are being contaminated by runoff from chlorine and bromine containing pesticides, raising concerns for drinking water and irrigation. At the same time, hospitals face shortages of iodine-based antiseptics, and local authorities plan to disinfect municipal water supplies to prevent outbreaks. Your supervisor has asked you to use periodic trends and halogen chemistry to assess risks and recommend safe, practical solutions that balance public health, agriculture, and environmental protection.

Available laboratory data:

| Element | Atomic Number | Electronegativity | Bond Energy (kJ $mol^{-1}$) |
|---|---|---|---|
| F | 9 | 3.98 | 158 |
| Cl | 17 | 3.16 | 243 |
| Br | 35 | 2.96 | 193 |
| I | 53 | 2.66 | 151 |

Chlorine also forms several oxoacids, including:

- Hypochlorous acid (HOCl) – used in water disinfection and bleaching.
- Chlorous acid ($HClO_3$) – used in herbicides and laboratory reagents.
- Perchloric acid ($HClO_4$) – used in rocket propellants and explosives.

**Tasks**

a) Use the data on electronegativity and bond energy to explain how the elements differ in chemical behavior and environmental impact.

b) Evaluate the risks of using compounds from these elements in water treatment and recommend safer alternatives to protect public health and the environment

**ITEM 4**: A rural community located near a mining and groundwater drilling site is experiencing unusual health problems of chronic fatigue, respiratory illness, and signs of radiation exposure. Preliminary investigations have revealed the presence of Radon-222, a radioactive noble gas produced from the natural decay of uranium in the surrounding bedrock. Water analysis has also confirmed elevated levels of Lead (a group 14 element likely leached from mineral ores), as well as Calcium and Chlorine compounds such hypochlorous acid (HOCl), used in paper bleaching and found in geological formations.

Table 1 below shows the comparison of the melting points of Lead with other group 14 elements:

| **Element** | **C** | **Si** | **Ge** | **Sn** | **Pb** |
|---|---|---|---|---|---|
| Melting point (°C) | 3500 | 1414 | 938 | 232 | 328 |

Local authorities have called upon your school's science team to conduct a chemical investigation that explains the situation in scientific terms. You are required to apply your knowledge of atomic structure, periodic properties, bonding, molecular behaviour, and radioactivity to analyze the contaminants and evaluate their implications for health, safety, and environmental management. Your goal is to explore and explain the underlying chemical concepts behind the contamination and offer insights that could inform safe water management strategies.

**Tasks**

a) Explain the chemical properties and environmental behaviour of the elements found in the bedrock.
b) Evaluate the health risks of the radioactive isotope and propose practical mitigation strategies.

**Part II: Attempt to any one item**

**ITEM 5:** In many rural Ugandan health centres, access to affordable and effective pain relief medication is limited due to the high cost of imported drugs. Some low-quality counterfeit painkillers on the market contain harmful additives and are poorly soluble in water, reducing their effectiveness and causing environmental harm when improperly disposed of. Concerned about this, a group of chemistry students is developing a painkiller inspired by paracetamol, a molecule with phenol and amine functional groups, as their project for exhibition during a science fair. However, they lack the knowledge of the structure, synthetic routes, mechanisms and how benzene can be used as a raw material in the process and have contacted you for help. They aim to create a painkiller that is effective and environmentally safe after use.

**Tasks**

As a chemistry learner,

a) Predict the solubility and acidity of the compound
b) Propose a synthetic route and a suitable mechanism for the first step in the production of the pain killer
c) Suggest suitable mitigations for the environmental risks of waste from production process.

**ITEM 6**: A certain town in central Uganda faces growing concerns over water pollution from agro-processing industries, especially during the cocoa harvest season. Wastewater from cocoa processing plants often contains organic matter including acids, and amines that can harm aquatic life if released untreated.

A group of entrepreneurs is setting up a small plant to process **cocoa husks**, a by-product of cocoa bean production, into value-added products. Chemical analysis of the husks shows that they contain ethanol (from natural fermentation), ethanoic acid, small amounts of amines (from protein breakdown), and aromatic aldehydes such as vanillin(4-hydroxy-3-methoxybenzaldehyde).
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3. React ethanoic acid with ethanol to produce an **ester** for fragrances, and study its reaction mechanism.
4. Evaluate the use of the esters and amines in producing fragrances and cosmetics

You have been tasked by your teacher to design feasible chemical processes, explain the underlying organic chemistry, and propose sustainable solutions to help the plant reduce environmental harm while maintaining profitability.

**Tasks**
As a chemistry learner, address the plans for the team and propose suitable solutions.